

# MongoDB Databases, Collections and Records

GitHub-readable preview of the presentation. Download the matching .pptx for the editable PowerPoint.

# 1. Syllabus tasks

- Create and drop databases and collections.
- Work with records using `find()`, `limit()`, `sort()`, `createIndex()` and `aggregate()`.

## 2. Workflow

### MongoDB Databases, Collections and Records



Workflow: input → processing → data/state → output

## 3.1. Code pathway — queries.js

```
1 // Database and collection operations.
2 use fsd2records;
3 db.createCollection('students');
4
5 db.students.insertMany([
6 { name: 'Rama', marks: 88, branch: 'CSE' },
7 { name: 'Gandhi', marks: 76, branch: 'CSE' },
8 { name: 'Roshni', marks: 92, branch: 'ECE' },
9 { name: 'Kiran', marks: 81, branch: 'CSE' }
10 ]);
```

## 3.2. Code pathway — queries.js

```
11
12 // FIND records.
13 db.students.find();
14
15 // LIMIT records.
16 db.students.find().limit(2);
17
18 // SORT by marks, descending.
19 db.students.find().sort({ marks: -1 });
20
```

## 4. Runtime walkthrough

### Runtime walkthrough — MongoDB Databases, Collections and Records



Runtime: start → enter → execute → observe

## 5. Run and observe

- Start the server / Vite app or open the MongoDB shell as described in the experiment README.
- Enter the sample input or call the required route.
- Watch the terminal for logs and errors.
- Verify the browser, API response, or database result.

## 6. Viva questions

- What is the purpose of this experiment?
- Which input is required?
- Which line or command performs the main operation?
- What output should you observe?
- What happens when the input changes?